

Questions for ch 3

1. **Proteins that are embedded within the cytoplasmic membrane and not easily extracted are called:**
 - ☐ a. internal
 - ☐ b. integral
 - ☐ c. external
 - ☐ d. peripheral
2. **When a substance is altered during transport across a membrane, it is called:**
 - ☐ a. Passive diffusion
 - ☐ b. Facilitated diffusion
 - ☐ c. Active transport
 - ☐ d. Group translocation
3. **Which of the following is not a major class of growth factors:**
 - ☐ a. Amino acids
 - ☐ b. Purines and pyrimidines
 - ☐ c. Vitamins
 - ☐ d. none of these
4. **Flexible bacteria with a helical shape are called:**
 - ☐ a. vibrios.
 - ☐ b. spirilla
 - ☐ c. spirochetes.
 - ☐ d. coccobacilli.
5. **The term Sarcineae denotes:**
 - ☐ a. grape-like clusters of staphylococcus
 - ☐ b. diplococcus or pairs of staphylococcus
 - ☐ c. 4 cocci in a chain
 - ☐ d. 8 cocci in a cubic form
6. **The steroid-like molecule that is believed to contribute to the formation of petroleum is**



called:

- ☐ a. the organic carbon found abundant in bacterial peptidoglycan layers
 - ☐ b. the squalene that is found in Archaeal membrane lipids
 - ☐ c. cholesterol that is abundant in bacterial cell walls
 - ☐ d. hopanoids that are abundant in bacterial membranes
7. The carrier proteins that are embedded in the membrane and increase the rate of diffusion of specific molecules across selectively permeable membranes are called:
- ☐ a. porins
 - ☐ b. Permeases
 - ☐ c. chaperones
 - ☐ d. peripheral proteins
8. One of the following is not a function of the cell wall:
- ☐ a. provides shape for the cell
 - ☐ b. provides protection against toxic substances
 - ☐ c. always contributes to the pathogenicity of the bacteria
 - ☐ d. protects the cell from osmotic lysis
9. The plasma membrane is the site of the following metabolic processes except:
- ☐ a. respiration
 - ☐ b. lipid synthesis
 - ☐ c. protein synthesis
 - ☐ d. synthesis of cell wall components
10. An open system in which the growth rate is maintained by the removal and addition of media at such a rate as to maintain a constant cell density is called a:
- ☐ a. manostat.
 - ☐ b. culturostat.
 - ☐ c. turbidostat.
 - ☐ d. chemostat.

Answers for Questions 1-10

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
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Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
b	d	d	c	d	d	b	c	c	d

Questions 11-20

11. All the following methods are used for estimation of viable bacterial cells except:

- o a. membrane filter method
- o b. spread plate technique
- o c. biomass determination
- o d. pour plate technique

12. During which phase are microorganisms most nearly uniform in terms of chemical and physiological properties?

- o a. decline
- o b. exponential
- o c. lag
- o d. stationary

13. Which of these methods can be used to determine the number of viable microorganisms in a sample?

- o a. counting a known volume of cells in a hemocytometer
- o b. light scattering in a spectrophotometer
- o c. measuring colony-forming units per ml
- o d. measuring total cell mass

14. Eosin methylene blue media supports the growth of:

- o a. both gram-positive and gram-negative
- o b. gram-negative
- o c. only supports Archaea
- o d. gram-positive

15. Addition of salt to a culture medium only allows the salt-tolerant bacteria to grow. This is an example of a:

- o a. Enriched media
- o b. Complex media



- o c. Selective media
- o d. Differential media

16. A prolonged decline in growth phase in a closed system results from:

- o a. the toxic waste that will kill the weak cells, leaving the strong ones.
- o b. the new media that is pumped through the culture.
- o c. the decrease in pH that will kill most of the organisms and only leave stronger ones.
- o d. nutrients released from dead cells that can be consumed by live cells.

17. A culture in a closed vessel to which no additional medium is added and from which no waste products are removed is called:

- o a. fed-batch culture
- o b. semicontinuous culture
- o c. batch culture
- o d. continuous culture

18. Which of the following procedures uses a photocell to measure absorbance of a culture to regulate the flow of culture media?

- o a. Turbidostat
- o b. Petroff-Hausser chamber
- o c. Coulter counter
- o d. Chemostat

19. Svedburg is a unit used to measure:

- o a. DNA/RNA size as a function of nucleotide length
- o b. protein size as a function of its sedimentation velocity
- o c. lipid size as a function of the hydrocarbon chain length
- o d. carbohydrate size as a function of polymer length

20. A plasmid that encodes the production of a toxin is called:

- o a. metabolic plasmid
- o b. virulence plasmid
- o c. col plasmid
- o d. Resistance plasmid



Answers for Questions 11-20

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
c	b	c	b	c	d	c	a	b	b

Questions 21-30

21. Chemotaxis is a process by which bacteria:

- o a. move on solid surfaces by means of type IV fimbriae.
- o b. move toward an attractant or away from a repellent.
- o c. respond metabolically to the presence of autoinducer molecules.
- o d. avoid phagocytosis.

22. Gram-positive cells retain the primary stain whereas gram-negative cells do not because:

- o a. the alcohol reduces the ability of the outer membrane to help trap the stain in gram-negative bacteria.
- o b. the alcohol reduces the ability of the outer membrane to help trap the stain in gram-negative bacteria and the alcohol shrinks the pores of the thick peptidoglycan layer of gram-positive cells, thereby trapping the stain.
- o c. the alcohol shrinks the pores of the thick peptidoglycan layer of gram-positive cells, thereby trapping the stain.
- o d. none of the choices.

23. Endospores are resistant to all the following except:

- o a. desiccation
- o b. heat and pressure at the same time
- o c. UV light radiation
- o d. chemical disinfectants

24. Many bacteria facilitate the uptake of iron by secreting low molecular weight molecules, called:

- o a. prions
- o b. siderophores
- o c. chaperones
- o d. permeases



25. Spores are resistant because of the following except:

- o a. the teichoic acid in the coat
- o b. dehydration of the coat
- o c. acid-soluble DNA binding proteins
- o d. dipicolinic acid in the coat

26. When the flagella is clustered at one or both ends, then this pattern is called:

- o a. peritrichous
- o b. monotrichous
- o c. lophotrichous
- o d. amphitrichous

27. When the cell wall is removed from a Gram-negative bacterium without removing the outer membrane, the resulting form is called:

- o a. spheroplast
- o b. trophoplast
- o c. protoplast
- o d. hopanoids

28. The most abundant protein(s) in the outer membrane of gram-negative bacteria is:

- o a. the porin proteins
- o b. glycoproteins
- o c. the peripheral proteins
- o d. The brawn's protein

29. Carboxysomes are:

- o a. inorganic inclusion bodies containing magnetosomes.
- o b. organic inclusion bodies containing CO₂-fixing enzymes.
- o c. vesicles containing enzymes for photosynthesis.
- o d. large polypeptides containing equal amounts of arginine and aspartic acid.

30. One of the following is not found in the bacterial cytoplasm:

- o a. ribosomes
- o b. inclusion bodies



- o c. actin protein
- o d. nucleoid

Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
b	b	b	b	a	c	a	a	b	c

31. Procaryotic nucleoids are made of the following:

- o a. 60% DNA; 30% proteins; 10% RNA
- o b. 50% DNA; 30% RNA; 20% Protein
- o c. 60% DNA; 30% RNA; 10% Protein
- o d. 60% DNA and 40% RNA

32. All the following are types of plasmids except:

- o a. resistance plasmids
- o b. bacteriocin encoding plasmids
- o c. conjugative plasmids
- o d. transposone plasmids

33. The basal body of a Gram-positive flagella consists of the following rings:

- o a. L ring and P ring
- o b. P ring and M ring
- o c. P ring and S ring
- o d. L ring, P ring, S ring, and M ring

34. The 70S prokaryotic ribosome consists of:

- o a. two 40S subunits
- o b. 50S subunit and 20S subunit
- o c. 40S subunit and 30S subunit
- o d. 50S subunit and 30S subunit

35. When a microorganism is placed in a _____ solution, water will enter into the cell and cause it to burst unless something is done to prevent this.

- o a. osmotic



- ☐ b. isotonic
 - ☐ c. hypertonic
 - ☐ d. hypotonic
- 36. The layer of protein or glycoprotein that exhibits a pattern resembling floor tiles on the surface of some microorganisms is called:
 - ☐ a. S layer
 - ☐ b. capsule
 - ☐ c. slime layer
 - ☐ d. cell wall
- 37. Which of the following is MOST effective against resistant endospores?
 - ☐ a. boiling
 - ☐ b. autoclaving
 - ☐ c. pasteurization
 - ☐ d. All of these are equally effective against resistant endospores.
- 38. The disinfectant of choice for municipal water supplies is:
 - ☐ a. either fluorine or chlorine
 - ☐ b. fluorine
 - ☐ c. chlorine
 - ☐ d. ultraviolet light
- 39. The prevention of infection caused by microorganisms is called:
 - ☐ a. sanitization.
 - ☐ b. sterilization.
 - ☐ c. antisepsis.
 - ☐ d. disinfection.
- 40. The reduction of the microbial population to levels that are considered safe by public health standards is called:
 - ☐ a. sanitization.
 - ☐ b. disinfection.
 - ☐ c. antisepsis.



- o d. sterilization.

Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
c	d	d	d	d	a	b	c	c	a

Questions 41-50

41. The increase in temperature needed to reduce the decimal reduction time to 10% of its value is the:

- o a. decimal reduction time (D value).
- o b. thermal death time (TDT).
- o c. z value.
- o d. thermal death point (TDP).

42. The shortest period of time needed to kill all organisms in a sample at a specified temperature is the:

- o a. thermal death time (TDT).
- o b. z value.
- o c. decimal reduction time (D value).
- o d. thermal death point (TDP).

43. To be effective as an antiseptic, alcohol concentration should be:

- o a. 70%
- o b. 85%
- o c. 100%
- o d. 95%

44. An agent that specifically kills fungi but not other kinds of microorganisms is also known as a:

- o a. fungicidal agent.
- o b. mycostatic agent.
- o c. germicidal agent.
- o d. fungistatic agent.

45. When killing microorganisms is done by dry heat, microbial death occurs by:



- ☐ a. denaturation of cell constituents only.
 - ☐ b. none of these.
 - ☐ c. denaturation of proteins and oxidation of cell constituents.
 - ☐ d. denaturation of cell constituents and oxidation of proteins.
- 46. Which of the following environmental factors generally do(es) not have an impact on the efficiency of an antimicrobial agent?
 - ☐ a. pH
 - ☐ b. concentration of organic matter
 - ☐ c. viscosity
 - ☐ d. refractive index
- 47. Which of the following items could be sterilized by dry heat sterilization?
 - ☐ a. intravenous (IV) solutions
 - ☐ b. glass pipets
 - ☐ c. rubber gloves
 - ☐ d. plastic petri dishes
- 48. Which of the following is normally used to remove microorganisms from air?
 - ☐ a. None of the above is correct.
 - ☐ b. HEPA filters
 - ☐ c. membrane filters
 - ☐ d. depth filters
- 49. Iodine can be complexed with an organic carrier to form water-soluble, stable complexes called:
 - ☐ a. iodochromes.
 - ☐ b. iodophors.
 - ☐ c. tinctures.
 - ☐ d. indidae.
- 50. The killing of vegetative forms of pathogenic microorganisms on inanimate objects is called:
 - ☐ a. antisepsis.
 - ☐ b. disinfection.



- o c. sanitization.
- o d. sterilization.

Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
c	a	a	a	c	d	b	b	b	b

